IAT New Test 14

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

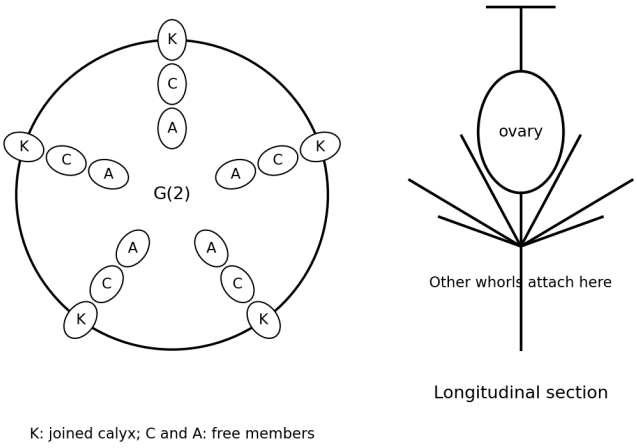
Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 A moss sporophyte remains attached to its leafy gametophyte. If the gametophyte is deprived of photosynthate while the capsule is still developing, which outcome is most directly predicted?
A) Capsule maturation may fail because sporophyte nutrition depends on the gametophyte
B) The sporophyte immediately becomes a free-living fern
C) The capsule changes its ploidy from diploid to haploid
D) Every spore becomes a gamete without meiosis

Q2 The diagram shows five united sepals, five free petals, five stamens and two fused carpels. Use the longitudinal section to infer ovary position. Which floral formula matches both views? Here brackets denote fusion.

Flower: whorls and longitudinal attachment



A) K5 C(5) A5 G(2) inferior
B) K(5) C5 A5 G(2) superior
C) K(5) C5 A(5) G2 superior
D) K5 C5 A5 G2 inferior

Q3 Cells exposed to a spindle poison replicate DNA normally but fail to attach some kinetochores. Which stage is expected to pause before sister chromatids separate?
A) G1 before S phase
B) Telophase after cytokinesis
C) Metaphase-to-anaphase transition
D) Prophase before chromosome condensation

Q4 In bright light with closed stomata, a C3 leaf shows net CO_2 uptake falling sharply, while a C4 leaf retains appreciable uptake at the same temperature. Which explanation best fits the contrast?
A) C4 plants never use RuBisCO
B) C3 plants cannot make ATP under bright light
C) C4 mesophyll lacks any oxygen
D) C4 CO_2 concentration in bundle-sheath cells suppresses RuBisCO oxygenation

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Q5 A heart pumps 5 L min^{-1} into a systemic circuit whose effective mean pressure drop is 100 mm Hg. If flow stays 5 L min^{-1} but vascular resistance falls by 20%, what pressure drop is required in the same linear-flow approximation?
A) 80 mm Hg
B) 100 mm Hg
C) 120 mm Hg
D) 64 mm Hg

Q6 After fertilisation, an ovule's integuments remain intact and the ovary wall expands. Which mature structures derive from these two tissues, in that order?
A) Endosperm and embryo
B) Seed coat and fruit wall
C) Embryo and seed coat
D) Fruit wall and endosperm

Q7 Two unaffected parents have a child with an autosomal recessive disorder. A second child is unaffected. If a new point mutation creates the same disease allele in one parent's germ line, which claim about the unaffected second child remains justified from the observed pedigree alone?
A) The child must be homozygous normal
B) The child must be a carrier
C) The child could be either homozygous normal or a carrier
D) The child's somatic cells must contain the new germ-line mutation

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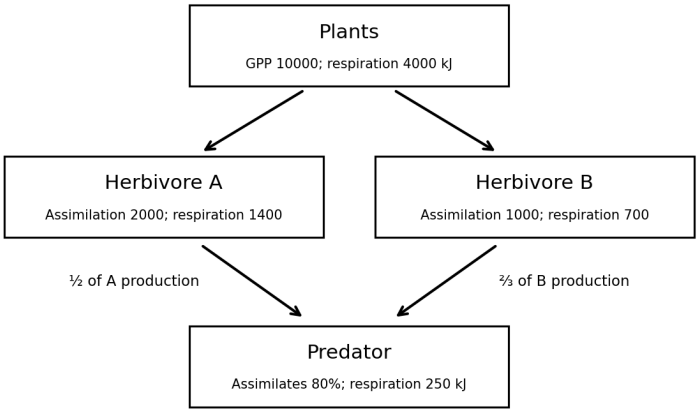
Q8 A virus infects host cells, and extracellular viral particles are neutralised by serum antibodies while already infected cells remain. Which additional response most directly removes those infected cells?
A) Plasma cells releasing more neutralising antibody
B) Complement acting only on free virions
C) Memory B cells without effector activation
D) Cytotoxic T cells recognising antigen displayed by infected cells

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Q9 A plasmid has intact marker S and a cloning site inside marker R. After ligation, colonies grow on S-antibiotic; a colony fails on R-antibiotic. Which control most directly tests whether failure is due to insertion at R rather than a nonrecombinant vector's ordinary phenotype?
A) Transform and plate an uncut parent vector on both antibiotics
B) Repeat only the same S plate without a vector control
C) Measure colony diameter without changing selection
D) Sequence an unrelated chromosomal gene

Q10 Plants have gross production 10000 kJ and respiration 4000 kJ over the measured interval. Herbivores A and B assimilate 2000 and 1000 kJ and respire 1400 and 700 kJ respectively. A predator consumes one half of A's new production and two thirds of B's, assimilates 80% of that food energy and respires 250 kJ. The diagram shows these routes. What fraction of plant net primary production becomes new predator biomass?

Trophic production and measured transfers



A) 1/15
B) 1/40
C) 1/12
D) 1/10

Q11 At a locus, measured allele frequencies are p=0.7 and q=0.3. Observed genotype frequencies are AA=0.60, Aa=0.20, aa=0.20. Which conclusion is supported?
A) The allele frequencies cannot be 0.7 and 0.3
B) The genotype frequencies exactly match Hardy–Weinberg proportions
C) The genotype frequencies depart from Hardy–Weinberg proportions despite matching the stated allele frequencies
D) Heterozygotes must have frequency 0.49 at equilibrium

Q12 After a pesticide is applied, a rare heritable resistant insect type becomes common. Which additional finding best distinguishes natural selection from an induced, directed resistance mutation in every survivor?
A) All survivors are identical in appearance
B) Resistance occurs only after exposure
C) Pesticide concentration was high
D) Resistance was detected in archived pre-exposure specimens

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Q13 At fixed tissue oxygen demand, a blood sample with lower pH releases more O_2 at the same partial pressure than a control. Which shift of the haemoglobin dissociation curve is consistent with that observation?
A) Left, higher affinity
B) No shift, unchanged affinity
C) Only a vertical shift in maximum haemoglobin amount
D) Right, lower affinity

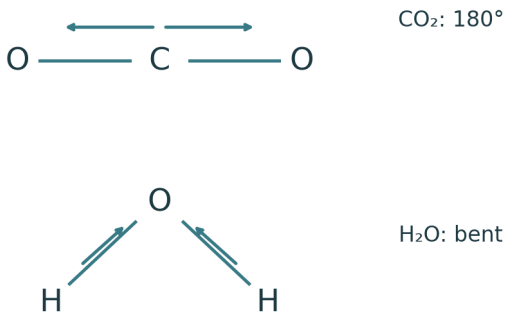
Q14 A disaccharide is hydrolysed to two monosaccharides without changing their carbon skeletons. Which bond was cleaved directly?
A) A glycosidic bond between sugar residues
B) A peptide bond between amino acids
C) A phosphodiester bond in nucleic acid
D) A disulfide bond between cysteines

Q15 Comparable surveys find 40 species in area A and 80 in area 16A, fitting $S = cA^z$. A restored region has area 4A, but its fitted habitat coefficient is only $0.75c$. Using this stated model, what is its species richness divided by the original richness in area A?
A) 3/4
B) $3\sqrt{2}/4$
C) 3/2
D) 2

Chemistry

Q16 The diagram places two identical polar C–O bonds symmetrically at 180° in CO_2, and two polar O–H bonds at an angle in H_2O. Which molecule has zero net electric dipole in its shown geometry?

Bond arrangement and bond dipoles



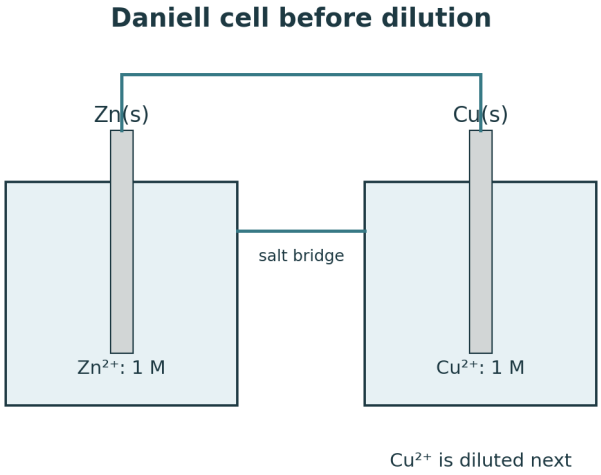
A) CO_2 only
B) H_2O only
C) Neither
D) Both

Q17 A cobalt compound has empirical composition $CoCl_3 \cdot 4NH_3$. Each mole gives one mole of immediate AgCl when treated with excess $AgNO_3$, and the dissolved complex gives two ions per formula unit. Assume coordinated chloride is not immediately displaced and the octahedral cobalt ion is low spin. Which formula, coordination number and unpaired-electron count fit?
A) $[Co(NH_3)_4Cl_2]Cl;\ 6;\ 0$
B) $[Co(NH_3)_4Cl]Cl_2;\ 5;\ 0$
C) $[Co(NH_3)_4]Cl_3;\ 4;\ 4$
D) $[Co(NH_3)_4Cl_2]Cl;\ 6;\ 4$

Q18 For a reaction with $\Delta H = -20$ kJ mol^{-1} and $\Delta S = -50$ J mol^{-1} K^{-1}, at what temperature range is ΔG negative under the constant-ΔH,ΔS approximation?
A) Above 400 K
B) Below 400 K
C) At all positive temperatures
D) At no positive temperature

Q19 Under base-catalysed aldol conditions, which carbonyl compound can form an enolate from its own α-hydrogen and undergo self-aldol addition?
A) Benzaldehyde
B) Formaldehyde
C) Propanal
D) Benzophenone

Q20 A Daniell cell is built with $Zn(s)|Zn^{2+}(1\ M)$ and $Cu^{2+}(1\ M)|Cu(s)$. Immediately after a small amount of Cu^{2+} solution is diluted without changing Zn^{2+} concentration, how does the reaction quotient $Q=[Zn^{2+}]/[Cu^{2+}]$ change and what is the direction of E relative to its earlier value?



A) Q falls, E rises
B) Q rises, E rises
C) Q falls, E falls
D) Q rises, E falls

Q21 An unknown amine gives a positive carbylamine test with $CHCl_3$ and alcoholic KOH. A separate sample gives a cold, relatively stable diazonium salt with $NaNO_2$/HCl at 0–5°C, and this couples with alkaline phenol. Which class best fits both observations?
A) Primary aromatic amine
B) Primary aliphatic amine
C) Secondary aromatic amine
D) Tertiary aromatic amine

Q22 In acidic solution, $3MnO_4^{2-}+4H^+\rightarrow 2MnO_4^-+MnO_2+2H_2O$ is proposed. Is this a disproportionation of Mn(VI), and which products hold its oxidation and reduction outcomes?
A) No, Mn remains +6 in all species
B) Yes, Mn(VII) in MnO_4^- and Mn(IV) in MnO_2
C) Yes, Mn(V) and Mn(VI) are products
D) No, oxygen alone undergoes redox

Q23 For a reaction with activation energy 50 kJ mol^{-1}, estimate $\ln(k_2/k_1)$ when temperature rises from 300 K to 330 K; use R=8.3 J mol^{-1} K^{-1} and constant activation energy.
A) −1.83
B) 0.18
C) 1.83
D) 18.3

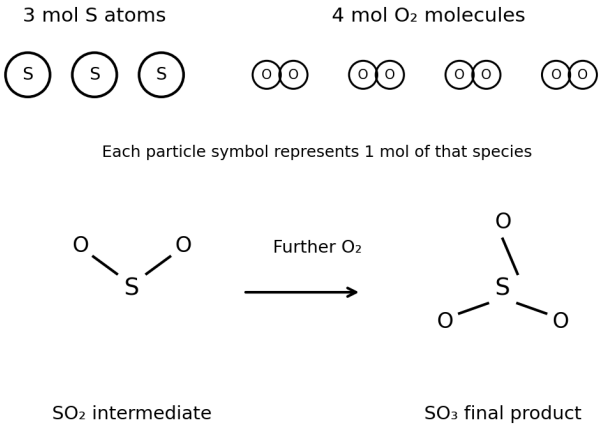
Q24 Acetyl chloride is added slowly to excess water. Which principal organic product forms after hydrolysis?
A) Ethanol
B) Ethanal
C) Acetic anhydride
D) Ethanoic acid

Q25 Two compounds have formula C_2H_6O: one is ethanol and the other dimethyl ether. Which relationship describes them?
A) Functional isomers
B) Geometrical isomers
C) Optical isomers
D) Conformers of one molecule

Q26 A solute forms only dimers in a dilute ideal solution. Its apparent molar mass inferred from a colligative measurement is 1.50 times its true monomer molar mass. What fraction of the original monomer units is associated into dimers?
A) 1/3
B) 2/3
C) 1
D) 4/3

Q27 A diagram represents 3 mol sulfur atoms and 4 mol O_2 molecules. Sulfur is converted through $S + O_2 \longrightarrow SO_2$ and $2\,SO_2 + O_2 \longrightarrow 2\,SO_3$. The two stages may be allocated to maximise SO_3 with no other reactions. What is the greatest possible SO_3 amount?

Initial particles and two oxidation stages



A) 3 mol
B) 2 mol
C) 8/3 mol
D) 4 mol

Q28 An element has successive ionisation energies I_1=580, I_2=1817, I_3=2745 and I_4=11578 kJ mol^{-1}. Which number of valence electrons and main-group number are consistent with the sharp jump?
A) One; group 1
B) Three; group 13
C) Two; group 2
D) Four; group 14

Q29 A DNA duplex region has 30 adenine bases and 20 guanine bases across both strands. How many thymine and cytosine bases, respectively, must it contain under ordinary Watson–Crick pairing?
A) 20 and 30
B) 30 and 30
C) 30 and 20
D) 20 and 20

Q30 A buffer contains 0.20 mol CH_3COOH and 0.10 mol CH_3COO^- in 1 L. Add 0.05 mol strong base with negligible volume change. What is the new acid-to-base mole ratio before using any logarithmic pH formula?
A) 4:1
B) 2:1
C) 3:1
D) 1:1

Mathematics

Q31 The circle $(x-2)^2+(y+1)^2=9$ touches which of the following horizontal lines?
A) $y=2$
B) $y=0$
C) $y=-1$
D) $y=4$

Q32 On the closed interval $[-1,2]$, what is the maximum of $f(x)=x^3-3x+1$?
A) 5
B) 1
C) 2
D) 3

Q33 A curve is given implicitly by $x^2y+y^3=2$. At its point (1,1), what is the tangent slope?
A) -2
B) $1/2$
C) 0
D) $-1/2$

Q34 Evaluate $\int_0^1 dx/[(x+1)^2(x+2)]$.
A) $\ln(4/3)$
B) $1/2-\ln(4/3)$
C) $1/2+\ln(4/3)$
D) $1/2$

Q35 Let $f(x)=\ln(x^2-1)$ for $x>1$, and let g be its inverse. Which pair gives $g(\ln 3)$ and the domain of g?
A) $-2;\ \mathbb{R}$
B) $2;\ (0,\infty)$
C) $2;\ \mathbb{R}$
D) $\sqrt{2};\ \mathbb{R}$

Q36 Vectors a=(1,1,0) and b=(1,−1,2) are viewed by an observer on the positive z-axis looking toward the origin. Which pair gives a×b and whether its z-component points into or out of the screen?
A) (−2,2,2), out of screen
B) (2,−2,2), out of screen
C) (2,−2,0), in the screen
D) (2,−2,−2), into the screen

Q37 Let $A=\begin{pmatrix} t & 1 \\ 1 & t \end{pmatrix}$, with real t. For which t does A have an inverse whose trace (sum of diagonal entries) is 1?
A) $1\pm\sqrt{2}$
B) $t=1$
C) $t=\pm 1$
D) $t=-1$

Q38 Two different tickets are drawn without replacement from five numbered 1 to 5, with unordered pairs equally likely. Given that their product is even, what is the probability both tickets are even?
A) 1/10
B) 1/7
C) 2/5
D) 1/2

Q39 How many nonempty multisets of size 4 can be chosen from 3 available types if supply is unlimited?
A) 12
B) 10
C) 15
D) 14

Q40 Let p be a positive integer. In the expansion of $(x^p+x^{-2})^6$, the coefficient of the term independent of x is 15. What is the sum of all possible p?
A) 1
B) 4
C) 6
D) 5

Q41 Three positive, strictly increasing numbers a, b and c form a geometric progression. Their sum is 13 and their product is 27. What is the sum of the first five terms of that progression beginning with a?
A) 121
B) 40
C) 120
D) 243

Q42 Polynomials x^2-3x+2 and x^2-kx+4 share one real root. What is the sum of all real possible k?
A) 5
B) 9
C) 7
D) 6

Q43 A data set has variance 9. If every value is replaced by 2x+5, what is the new variance?
A) 6
B) 18
C) 36
D) 49

Q44 A one-parameter family of curves is $y=Cx^2$ for $x>0$. Eliminate C to form its differential equation.
A) $y'=Cx$
B) $xy'=y$
C) $y'=2y/x$
D) $y'=2x/y$

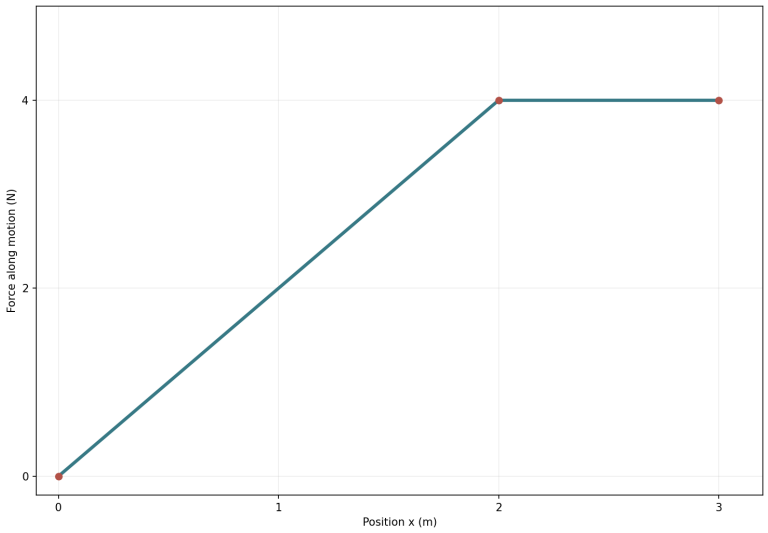
Q45 Evaluate $\lim_{x\to 0} (e^{(2x)}-1-2x)/x^2$.
A) 2
B) 1
C) 4
D) 0

Physics

Q46 A ball is thrown vertically upward at 20 m s^{-1} from a balcony; take g=10 m s^{-2}. At what time is its height above the balcony greatest?
A) 1 s
B) 2 s
C) 3 s
D) 4 s

Q47 An elevator accelerates upward at 2 m s^{-2}. A 3 kg mass hangs stationary relative to its cabin from a light cord. What tension does the cord exert? Use g=10 m s^{-2}.
A) 24 N
B) 30 N
C) 36 N
D) 42 N

Q48 A 2 kg particle enters x=0 at 2 m s^{-1}. The pictured force along its motion is F(x)=2x N from 0 to 2 m, then 4 N from 2 to 3 m. With no other work, what speed does it have at x=3 m?



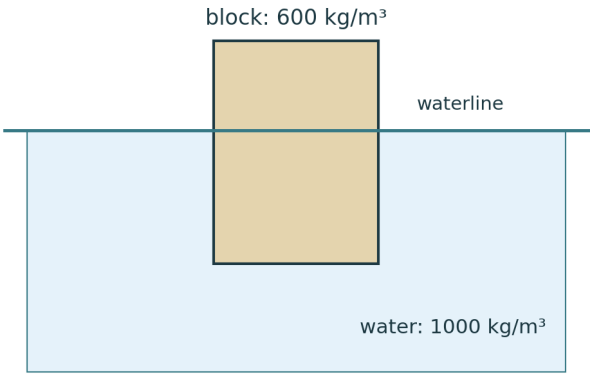
A) 2 m s^{-1}
B) √8 m s^{-1}
C) 4 m s^{-1}
D) 2√3 m s^{-1}

Q49 A uniform solid disk rolls without slipping down from rest through vertical height h. What fraction of its final kinetic energy is rotational about its centre?
A) 1/3
B) 1/4
C) 1/2
D) 2/3

Q50 A planet has twice Earth's mass and twice Earth's radius. What is its surface escape speed divided by Earth's, neglecting rotation?
A) 1/√2
B) 1
C) √2
D) 2

Q51 A uniform block of density 600 kg m⁻³ floats upright in water of density 1000 kg m⁻³. What fraction of its height is submerged at equilibrium?

Block floating at equilibrium



A) 0.4
B) 0.5
C) 0.6
D) 1.0

Q52 A gas absorbs 50 J heat and does 20 J work on surroundings. Under $\Delta U = Q - W$, what is its internal-energy change?
A) −70 J
B) −30 J
C) 70 J
D) 30 J

Q53 Two identical positive charges +q are fixed at x=−a and x=+a. At which finite x on their joining axis is the net electric field zero?

Equal positive charges on one axis



A) x=0
B) x=−2a
C) x=2a
D) No finite point

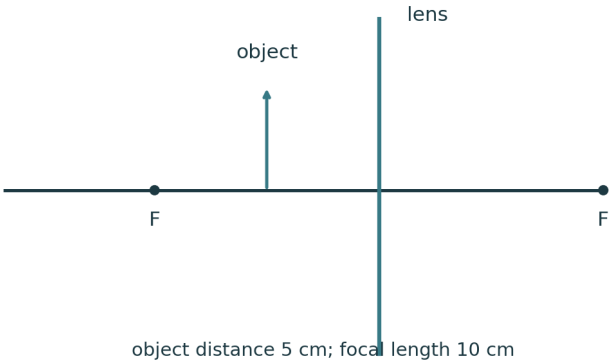
Q54 A capacitor is charged and then disconnected from its battery. A dielectric of relative permittivity 4 completely fills the gap. What is final-to-initial electrostatic energy ratio?
A) 4
B) 1/4
C) 1/16
D) 16

Q55 A charge +q moves at v=2 m s^{-1} along +x in a magnetic field B=3 T along +y. What is magnetic-force direction and magnitude, for q=1 C?
A) −z, 6 N
B) +z, 3 N
C) +z, 6 N
D) −y, 6 N

Q56 An ideal inductor of 2 H carries 3 A. What energy is stored in its magnetic field?
A) 3 J
B) 6 J
C) 12 J
D) 9 J

Q57 A convex lens with f=10 cm has an object 5 cm in front of it. Which image is formed, using Cartesian signs with light left to right?

Converging lens and near object



A) Virtual upright on object side, 10 cm from lens
B) Real inverted on far side, 10 cm
C) Virtual inverted on far side, 5 cm
D) Real upright on object side, 5 cm

Q58 In a single-slit Fraunhofer setup, slit width is halved while wavelength and screen distance remain fixed. What happens to the central maximum's linear width?
A) Halves
B) Doubles
C) Quadruples
D) Unchanged

Q59 Which spectral feature distinguishes a hot dilute atomic gas from a hot dense source viewed through a dispersing prism?
A) Continuous spectrum in all cases
B) No radiation from dilute gas
C) Bright discrete emission lines at characteristic wavelengths
D) Only a single universal line for all elements

Q60 A detector reads 900 counts min^{-1} at t=0 and 500 counts min^{-1} at t=4 h from one radioactive source plus a constant electronic background of 100 counts min^{-1}. Immediately before the reading at t=12 h, a screen is inserted that halves detected source events but does not change the electronic background. What total rate is then expected?
A) 100 counts min^{-1}
B) 200 counts min^{-1}
C) 250 counts min^{-1}
D) 150 counts min^{-1}

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / A bryophyte sporophyte dependence

Answer: (A) The developing capsule may lack supply.

Why: In bryophytes the sporophyte is attached to and nutritionally dependent on the dominant gametophyte, so removing that supply threatens its growth. The capsule still contains diploid sporogenous tissue until meiosis produces haploid spores; starvation does not rewrite that life-cycle sequence.

Why not the others: (B) changes the taxon; (C) confuses nutritional state with chromosome number; (D) replaces meiotic spore formation with gamete production.

Key point: Distinguish dependence from ploidy.

Q2 / B floral formula inference

Answer: (B) K(5) C5 A5 G(2), superior.

Why: The calyx is fused, petals and stamens are free, and the bicarpellary gynoecium is syncarpous. The diagram places other whorls below the ovary, supporting the superior assignment; no fusion of stamens is indicated.

Why not the others: (A) swaps sepal/petal fusion and ovary position; (C) invents stamen fusion and free carpels; (D) loses both fusions and reverses ovary position.

Key point: Translate each depicted whorl independently.

Q3 / C mitotic checkpoint

Answer: (C) The spindle checkpoint delays anaphase.

Why: Unattached kinetochores signal that chromosome segregation is not yet safe, preventing the transition that separates sister chromatids. DNA replication already occurred, so an S-phase arrest does not explain the observed attachment defect.

Why not the others: (A) is earlier than replication; (B) is after segregation; (D) concerns condensation rather than attachment.

Key point: A checkpoint responds to a specific unfinished process.

Q4 / D photorespiration contrast

Answer: (D) Spatial concentration suppresses photorespiration.

Why: Closing stomata lowers internal CO_2 and favours RuBisCO oxygenation in C3 leaves, reducing net fixation. C4 fixation and transfer elevate CO_2 near bundle-sheath RuBisCO, so the same external limitation has a smaller photorespiratory penalty.

Why not the others: (A) C4 bundle sheaths use RuBisCO; (B) light reactions persist; (C) oxygen is present in C4 tissues.

Key point: Compare local RuBisCO environments.

Q5 / A cardiac output and pressure

Answer: (A) 80 mm Hg.

Given: Pressure drop is flow times effective resistance in the stipulated model.

Formula/Principle: $\Delta P = QR$; Q unchanged, $R_{new} = 0.8 R_{old}$.

Step 1: $\Delta P_{new} = 0.8 \times 100 = 80$ mm Hg.

Why: Holding cardiac output fixed means the pressure gradient scales with the resistance, not with stroke volume or heart rate separately. This is a simplified mean-flow model; it does not assert that every instantaneous arterial pressure follows the same ratio.

Why not the others: (B) ignores changed R; (C) raises instead of lowers it; (D) incorrectly squares the factor.

Key point: Keep the held quantity explicit.

Q6 / B seed and fruit development

Answer: (B) Seed coat, fruit wall.

Why: The integuments surrounding the ovule become the protective seed coat, while the ovary wall develops into the fruit wall or pericarp. Endosperm and embryo arise from fertilisation products, not directly from either of the named maternal protective layers.

Why not the others: (A) names fertilisation products; (C) assigns embryo to integuments; (D) reverses tissue origin and inserts endosperm.

Key point: Trace maternal tissues and fusion products separately.

Q7 / C mutation and disorder

Answer: (C) The unaffected genotype is not uniquely fixed.

Why: Under recessive inheritance, unaffected status permits both AA and Aa, even when parental transmission can be complicated by a new germ-line mutation. A mutation arising in a parent's germ line is not automatically present in that parent's somatic cells or in every sibling.

Why not the others: (A) excludes carriers without evidence; (B) treats possible carrier status as certain; (D) forces inheritance of an event from another gamete.

Key point: Distinguish phenotype from an unobserved genotype.

Q8 / D antibody versus cellular defence

Answer: (D) Cytotoxic T-cell effector response.

Why: Antibody can block extracellular entry but cannot by itself reach viral proteins inside an infected cell. Cytotoxic T cells recognise peptides presented on infected-cell MHC I and can kill the cellular reservoir, addressing the residual infection described.

Why not the others: (A) acts chiefly on extracellular particles; (B) does not target infected intracellular reservoirs as stated; (C) is a memory state rather than immediate cellular killing.

Key point: Match the pathogen's location to defence mechanism.

Q9 / A recombinant selection control

Answer: (A) Compare intact parent-vector phenotype.

Why: An uncut plasmid should retain both markers, establishing that R-antibiotic growth is possible for the parent vector under the same conditions. If the recombinant colony alone loses R growth while retaining S, insertional disruption becomes more credible than a baseline selection failure.

Why not the others: (B) does not test R baseline; (C) does not assay marker function; (D) misses the plasmid locus.

Key point: A control must preserve the causal feature being tested.

Q10 / B Multi-compartment trophic energy accounting

Answer: (B) 1/40.

Given: Each stated respiration loss belongs to its own trophic compartment; unconsumed production remains outside the predator's budget.

Formula/Principle: Net production equals assimilated energy minus respiration at each compartment.

Step 1:

$$\text{NPP} = 10000 - 4000 = 6000 \text{ kJ}.$$

Step 2:

$$P_A = 2000 - 1400 = 600, \quad P_B = 1000 - 700 = 300 \text{ kJ}.$$

Step 3: Predator food energy is

$$(1/2)(600) + (2/3)(300) = 500 \text{ kJ}.$$

Step 4: Predator new biomass energy is

$$P_P = 0.80(500) - 250 = 150 \text{ kJ}.$$

Step 5:

$$P_P/\text{NPP} = 150/6000 = 1/40.$$

Why: Consumption, assimilation and net production are different stages of the transfer budget. Subtracting respiration at the right stage prevents plant gross production or undigested food from being counted as predator growth.

Why not the others: (A) uses predator assimilation before its respiration; (C) uses consumed food before assimilation and respiration; (D) imports a generic ten-percent rule instead of the supplied measurements.

Key point: Follow each trophic compartment's mass-energy budget before forming an efficiency ratio.

Q11 / C Hardy-Weinberg assumption

Answer: (C) Frequencies deviate from equilibrium.

Given: Observed A allele frequency=0.60+0.20/2=0.70.

Formula/Principle: Equilibrium expected p^2,2pq,q^2=0.49,0.42,0.09.

Step 1: Compare expected to 0.60,0.20,0.20; all differ.

Why: The allele frequencies can be correct while genotype proportions depart from random-mating equilibrium. The result alone does not identify which assumption failed, such as selection, structure or nonrandom mating.

Why not the others: (A) ignores observed allele calculation; (B) overlooks expected 0.42 heterozygotes; (D) confuses p^2 with 2pq.

Key point: Test genotype proportions, not just p+q=1.

Q12 / D evolutionary selection evidence

Answer: (D) Earlier occurrence supports selection.

Why: Finding the resistance phenotype before exposure establishes that a heritable variant existed for the pesticide to select. The observed increase after exposure can then be explained by differential survival and reproduction without assuming the chemical produced the needed mutation on demand.

Why not the others: (A) appearance may hide genotype; (B) would not establish pre-existing variation; (C) says little about origin.

Key point: Temporal evidence separates variation from selection.

Q13 / D gas transport and dissociation

Answer: (D) Right shift and greater unloading.

Why: Higher H^+ concentration stabilises lower-affinity haemoglobin states, so the same oxygen partial pressure gives lower saturation and more release to tissues. The curve shift concerns binding equilibrium, not a change in the number of haemoglobin molecules in the sample.

Why not the others: (A) predicts less unloading; (B) contradicts measured release; (C) changes capacity rather than affinity.

Key point: Read saturation at fixed partial pressure.

Q14 / A biomolecule linkage

Answer: (A) Glycosidic linkage.

Why: Condensation joins monosaccharides through an O-glycosidic linkage, and hydrolysis adds water across that linkage to release the original sugar units. The other named bonds connect different biomolecular building blocks and do not define a disaccharide.

Why not the others: (B) joins amino acids; (C) joins nucleotides; (D) stabilises proteins.

Key point: Identify linkage from monomer identity.

Q15 / B Species-area exponent and changed habitat coefficient

Answer: (B) $3\sqrt{2}/4$.

Given: The initial two surveys share c; the restored region has a separately stipulated coefficient $c' = 0.75c$.

Formula/Principle: Infer z from the comparable surveys before combining area and coefficient changes.

Step 1:

$$\frac{80}{40} = 16^z \Rightarrow 2 = 16^z \Rightarrow z = \frac{1}{4}.$$

Step 2:

$$\frac{S_{\text{restored}}}{S_A} = \frac{0.75c(4A)^{1/4}}{cA^{1/4}} = 0.75\,4^{1/4}.$$

Step 3:

$$S_{\text{restored}}/S_A = \frac{3}{4}\sqrt{2} = \frac{3\sqrt{2}}{4}.$$

Why: The original paired surveys allow c to cancel only when estimating the exponent. In the restoration comparison the habitat coefficient changes, so cancelling it as if it were unchanged would overestimate richness.

Why not the others: (A) includes coefficient loss but ignores added area; (C) uses an area exponent one half; (D) ignores the coefficient change and uses the wrong area scaling.

Key point: Infer an exponent using comparable habitats, then retain any specified coefficient change.

Chemistry

Q16 / A bond polarity versus molecular dipole

Answer: (A) CO_2 only.

Why: Its equal bond-dipole vectors oppose along one line and cancel; the bent water vectors have a nonzero resultant toward oxygen. Polar bonds therefore do not guarantee a polar molecule unless their vector arrangement is considered.

Why not the others: (B) reverses the geometries; (C) ignores cancellation in linear CO_2; (D) ignores water's bend.

Key point: Sum bond dipoles as vectors.

Q17 / A Werner formula inference and low-spin electron filling

Answer: (A) $[Co(NH_3)_4Cl_2]Cl$, coordination number six, zero unpaired electrons.

Given: Exactly one chloride is a free counterion; four ammonia molecules are neutral donors.

Formula/Principle: Combine free-ion evidence, coordination count, oxidation state and low-spin d-electron filling.

Step 1: One free chloride leaves two chloride ligands inside the sphere, giving $[Co(NH_3)_4Cl_2]^+$ and one Cl^-.

Step 2: Four N donors plus two Cl donors give coordination number six. Cobalt satisfies $x - 2 = +1$, so $x = +3$.

Step 3: Co(III) is d^6; low-spin octahedral filling is $t_{2g}^6 e_g^0$, with no unpaired electrons.

Worked derivation:

$$N_{\text{free Cl}} = 1, \quad x - 2 = +1 \Rightarrow x = +3, \quad \text{CN} = 4 + 2 = 6, \quad d^6_{\text{low spin}} : t_{2g}^6 e_g^0.$$

Why: Precipitation and ion count identify the sphere boundary before the electronic argument begins. The low-spin assumption is explicitly supplied; it must not be inferred merely from the number of ligands.

Why not the others: (B) predicts two free chlorides and three ions; (C) predicts three free chlorides and four ions; (D) uses high-spin d^6 filling despite the stated low-spin condition.

Key point: Establish the coordination formula experimentally before assigning electron filling.

Q18 / B Gibbs-temperature boundary

Answer: (B) Below 400 K.

Why: Converting entropy to -0.050 kJ mol^{-1} K^{-1} gives $\Delta G = -20 + 0.050T$, which crosses zero at 400 K. At lower temperatures the favourable exothermic term dominates the unfavourable entropy decrease; above it the sign reverses.

Why not the others: (A) reverses inequality; (C) ignores entropy; (D) ignores enthalpy.

Key point: Convert units before solving the temperature boundary.

Q19 / C aldol eligibility

Answer: (C) Propanal.

Why: Propanal has an α-carbon bearing hydrogens next to its carbonyl, permitting enolate formation and attack on another propanal carbonyl. Benzaldehyde, formaldehyde and benzophenone lack an enolisable α-hydrogen for their own enolate under the stated mechanism.

Why not the others: (A) benzylic ring carbon is not an α-CH; (B) has no α-carbon; (D) has phenyl groups but no α-H.

Key point: Locate α-hydrogen on carbon next to carbonyl.

Q20 / D cell polarity from quotient

Answer: (D) Q rises and E falls.

Why: Zinc oxidation produces Zn^{2+} while copper reduction consumes Cu^{2+}; lowering the copper reactant concentration raises the quotient. By the Nernst form $E=E°-(RT/2F)\ln Q$, the increased Q lowers the potential at fixed temperature.

Why not the others: (A) reverses quotient; (B) reverses Nernst sign; (C) reverses both.

Key point: Write Q from the balanced cell reaction.

Q21 / A paired amine class and aromaticity tests

Answer: (A) Primary aromatic amine.

Why: A positive carbylamine test establishes a primary amine, while a diazonium salt that persists cold and couples with phenol points to an aryl primary amine. Primary aliphatic diazonium species decompose rapidly instead of giving the stated stable coupling response.

Why not the others: (B) can give carbylamine but fails the persistent diazonium observation; (C) is not primary and cannot satisfy both tests; (D) likewise fails the primary-amine requirement.

Key point: Combine independent class and aryl-group evidence.

Q22 / B disproportionation assessment

Answer: (B) Mn(VI) gives +7 and +4.

Why: Mn in MnO_4^{2-} is +6; in MnO_4^- it is +7 and in MnO_2 it is +4. One portion of the same initial oxidation state loses electrons while another gains them, and the stated stoichiometry balances the electron exchange.

Why not the others: (A) misassigns product states; (C) invents Mn(V); (D) overlooks manganese changes.

Key point: Determine oxidation numbers for each species.

Q23 / C Arrhenius temperature comparison

Answer: (C) About 1.83.

Given: $\ln(k_2/k_1)=Ea/R(1/T_1-1/T_2)$.

Formula/Principle: 50000/8.3 ×30/(300×330).

Step 1: 6024×0.000303≈1.83.

Worked derivation:

$$\ln(k_2/k_1) = \frac{50000}{8.3}\left(\frac{1}{300}-\frac{1}{330}\right) \approx 1.83.$$

Why: Increasing temperature makes 1/T fall, so the reciprocal difference is positive and rate constant increases. The logarithm compares rates through activation energy; it is not directly the fractional temperature increase.

Why not the others: (A) reverses temperatures; (B) misses a factor ten; (D) adds one.

Key point: Keep Ea and R in matching joule units.

Q24 / D acid derivative transformation

Answer: (D) Ethanoic acid.

Why: Water attacks the reactive acyl carbon and chloride leaves, converting CH_3COCl to CH_3COOH with HCl as a coproduct. Excess water drives hydrolysis rather than condensation to anhydride.

Why not the others: (A) needs reduction; (B) needs selective reduction; (C) forms under dehydrating acylation conditions.

Key point: Replace an acyl leaving group with hydroxyl under hydrolysis.

Q25 / A isomer classification

Answer: (A) Functional isomers.

Why: Their atoms count identically, but ethanol contains an O–H alcohol group whereas dimethyl ether has C–O–C connectivity. This difference in functional group is structural isomerism, not a different orientation of the same bond framework.

Why not the others: (B) needs restricted rotation geometry; (C) needs chirality; (D) preserves connectivity.

Key point: Same formula can encode different functional groups.

Q26 / B Dimer association inferred from apparent molar mass

Answer: (B) 2/3.

Given: Let alpha be the fraction of original monomer units that associate; two such units make one dimer.

Formula/Principle: Particle factor $i = (1-\alpha) + \alpha/2 = 1 - \alpha/2$, while $M_{\text{apparent}} = M/i$.

Step 1:

$$i = \frac{M}{1.50M} = \frac{2}{3}.$$

Step 2:

$$1 - \frac{\alpha}{2} = \frac{2}{3} \Rightarrow \alpha = \frac{2}{3}.$$

Step 3: This lies between zero and one, as required for an associated fraction.

Why: Association reduces the number of independently moving solute particles, causing a colligative method to overestimate molar mass. Counting dimers as if each contained one original unit would violate the monomer-unit balance.

Why not the others: (A) reports $1 - i$ without the dimer factor; (C) would give a twofold apparent molar mass; (D) exceeds the physically allowed fraction.

Key point: Relate apparent molar mass to particle count before calculating association.

Q27 / C Oxygen allocation across two sulfur-oxidation stages

Answer: (C) $8/3$ mol.

Given: Initial sulfur is 3 mol and oxygen is 4 mol; the diagram distinguishes atomic sulfur from diatomic oxygen.

Formula/Principle: Adding the stages gives

$$2\,S + 3\,O_2 \longrightarrow 2\,SO_3.$$

Step 1: Each mole of SO_3 requires one mole S and $3/2$ mol O_2.

Step 2:

$$n_{SO_3} \leq \min\left(3, \frac{4}{3/2}\right) = \frac{8}{3}\ \text{mol}.$$

Step 3: This bound is attainable: $8/3$ mol O_2 makes $8/3$ mol SO_2, and the remaining $4/3$ mol O_2 oxidises it to SO_3.

Why: Oxygen is shared between the two stages, so using it entirely in the first stage need not maximise the final product. The achievable allocation confirms that the overall-reaction upper bound is not merely formal.

Why not the others: (A) ignores the oxygen shortage for complete final oxidation; (B) imposes an unnecessary integer-mole rounding; (D) treats each oxygen molecule as one product molecule without conserving sulfur.

Key point: Combine staged stoichiometry and verify an attainable reagent allocation.

Q28 / B successive ionisation-energy jump and valence group

Answer: (B) Three valence electrons; group 13.

Why: The exceptionally large increase from the third to the fourth removal means the first three came from the outer shell, while the fourth would remove an inner-shell electron. A main-group atom with three outer electrons belongs to group 13 under modern numbering; the absolute values are not needed to identify that boundary.

Why not the others: (A) puts the jump after the first electron; (C) puts it after the second; (D) treats the costly core electron as valence.

Key point: Locate the largest successive-energy jump before assigning group.

Q29 / C nucleic-acid base pairing

Answer: (C) 30 thymine, 20 cytosine.

Why: Each A pairs with one T, and each G pairs with one C, so duplex totals satisfy A=T and G=C. The counts are across both strands; assigning complementary counts per individual strand is unnecessary.

Why not the others: (A) swaps complementary partners; (B) overcounts C; (D) undercounts T.

Key point: Use complementary base totals in a duplex.

Q30 / D buffer before and after neutralisation

Answer: (D) 1:1.

Given: OH^- consumes HA, producing A^-.

Formula/Principle: nHA=0.20−0.05=0.15; nA^-=0.10+0.05=0.15.

Step 1: Ratio 0.15:0.15=1:1.

Worked derivation:

$$n_{\mathrm{HA}} = 0.20 - 0.05 = 0.15, \quad n_{\mathrm{A}^-} = 0.10 + 0.05 = 0.15, \quad n_{\mathrm{HA}} : n_{\mathrm{A}^-} = 1 : 1.$$

Why: Neutralisation occurs before buffer equilibrium is recalculated, and both members change in opposite directions by the same stoichiometric amount. Using the original ratio would ignore the chemical reaction with added base.

Why not the others: (A) reverses addition; (B) keeps original ratio; (C) changes only the numerator incorrectly.

Key point: Update mole counts before Henderson–Hasselbalch.

Mathematics

Q31 / A circle tangency

Answer: (A) y=2.

Given: Centre (2,−1), radius 3.

Formula/Principle: A horizontal tangent is at centre y plus or minus radius.

Step 1: y=−1±3 gives 2 and −4; only 2 is listed.

Worked derivation:

$$d((2,-1),y=c)=|c+1|=3\Rightarrow c=2 \text{ or } -4.$$

Why: Tangency means the perpendicular distance from centre to line equals radius. It is not enough that a line merely crosses the circle at two points.

Why not the others: (B) lies one unit away; (C) passes through centre; (D) lies five units away.

Key point: Use centre-to-line distance.

Q32 / D extreme value with endpoints

Answer: (D) 3.

Given: f′=3(x²−1); the interval endpoints are −1 and 2, with interior critical point 1.

Formula/Principle: Compare the function at every endpoint and interior critical point.

Step 1: f(−1)=3, f(1)=−1, f(2)=3, so the maximum is 3.

Worked derivation:

$$\max_{[-1,2]} f=\max\{f(-1),f(1),f(2)\}=\max\{3,-1,3\}=3.$$

Why: A closed interval requires both endpoints to be checked, and the interior stationary point is a minimum here. The maximum occurs at two different endpoints, but it has one numerical value.

Why not the others: (A) comes from overestimating an endpoint; (B) reads the constant term only; (C) omits the endpoint comparison.

Key point: Compare endpoints and critical points.

Q33 / D implicit tangent

Answer: (D) −1/2.

Given: The point satisfies 1²(1)+1³=2.

Formula/Principle: Implicit differentiation of x²y requires the product rule, and differentiation of y³ requires the chain rule.

Step 1: 2xy+(x²+3y²)y′=0; substitute (1,1) to obtain 2+4y′=0.

Worked derivation:

$$2xy+x^2y'+3y^2y'=0\Rightarrow y'=-\frac{2xy}{x^2+3y^2}\Rightarrow y'(1,1)=-\frac{1}{2}.$$

Why: Both the coefficient of y′ in x²y and the derivative of y³ contribute to the denominator. Evaluating the derivative only after differentiating the full implicit relation preserves the tangent direction.

Why not the others: (A) omits the y′ terms from y³; (B) drops the negative sign; (C) treats y as constant.

Key point: Differentiate both products and powers of y.

Q34 / B repeated-pole partial fractions with limits

Answer: (B) $1/2 - \ln(4/3)$.

Given: The repeated factor $(x+1)^2$ requires both first- and second-power partial fractions.

Formula/Principle:

$$\frac{1}{(x+1)^2(x+2)} = -\frac{1}{x+1} + \frac{1}{(x+1)^2} + \frac{1}{x+2}.$$

Step 1: Integrate each term:

$$F(x) = -\ln(x+1) - \frac{1}{x+1} + \ln(x+2).$$

Step 2:

$$F(1) = \ln(3/2) - 1/2, \qquad F(0) = \ln 2 - 1.$$

Step 3:

$$I = F(1) - F(0) = \frac{1}{2} + \ln(3/4) = \frac{1}{2} - \ln(4/3).$$

Worked derivation:

$$I = \frac{1}{2} - \ln\frac{4}{3} > 0.$$

Why: A repeated pole contributes a rational primitive as well as logarithms. Subtracting both lower-limit contributions is essential, and the positive final value is consistent with the positive integrand on the interval.

Why not the others: (A) omits the repeated-pole contribution; (C) reverses the logarithmic end-point subtraction; (D) cancels logarithms that have different arguments.

Key point: Include every repeated-factor power and evaluate both endpoints.

Q35 / C Inverse domain and branch of a logarithmic composition

Answer: (C) $2;\ \mathbb{R}$.

Given: For $x > 1$, $x^2 - 1$ takes every positive value, so its logarithm takes every real value.

Formula/Principle: Solve $y = \ln(x^2 - 1)$, retaining the positive original branch; the inverse domain is the original range.

Step 1:

$$e^y = x^2 - 1 \Rightarrow x^2 = e^y + 1.$$

Step 2: Since $x > 1$,

$$g(y) = \sqrt{e^y + 1}, \quad y \in \mathbb{R}.$$

Step 3:

$$g(\ln 3) = \sqrt{3+1} = 2.$$

Worked derivation:

$$g(y) = \sqrt{e^y + 1} > 1 \text{ for every real } y, \qquad g(\ln 3) = 2.$$

Why: Exponentiation undoes the logarithm but leaves two algebraic square-root branches; the original domain selects only the positive one. The full real inverse domain is valid because the exponential is positive for every real y, making the reconstructed input strictly greater than one.

Why not the others: (A) returns an excluded negative input; (B) incorrectly transfers positivity of the logarithm's argument to its output; (D) omits the added one after exponentiation.

Key point: Undo a composition in reverse order while preserving its branch and range.

Q36 / D cross-product orientation

Answer: (D) $(2, -2, -2)$, with its z-component into the screen.

Given: $\mathbf{a} = (1, 1, 0)$, $\mathbf{b} = (1, -1, 2)$; the observer is on the positive z-axis looking toward the origin.

Formula/Principle: Use the ordered cross product; positive z points toward that observer.

Step 1:

$$c_x = a_y b_z - a_z b_y = 1(2) - 0(-1) = 2.$$

Step 2:

$$c_y = a_z b_x - a_x b_z = 0(1) - 1(2) = -2, \quad c_z = a_x b_y - a_y b_x = 1(-1) - 1(1) = -2.$$

Step 3: The negative z-component points away from the observer and therefore into the screen.

Step 4: Verify perpendicularity:

$$\mathbf{a} \cdot \mathbf{c} = 2 - 2 = 0, \qquad \mathbf{b} \cdot \mathbf{c} = 2 + 2 - 4 = 0.$$

Worked derivation:

$$\mathbf{a} \times \mathbf{b} = 2\mathbf{i} - 2\mathbf{j} - 2\mathbf{k}.$$

Why: The vector must be perpendicular to both inputs, while its sign is fixed by their order. The viewing convention is a separate geometric check on the z-component, not a replacement for calculating all three components.

Why not the others: (A) reverses the cross-product order; (B) changes the z sign; (C) incorrectly drops the nonzero z-component.

Key point: Verify components, orthogonality and viewing orientation separately.

Q37 / A Matrix inverse with a trace constraint

Answer: (A) $1 \pm \sqrt{2}$.

Given: $\det A = t^2 - 1$, so $t \neq \pm 1$ is necessary.

Formula/Principle: Use the explicit inverse before imposing the trace condition.

Step 1:

$$A^{-1} = \frac{1}{t^2 - 1} \begin{pmatrix} t & -1 \\ -1 & t \end{pmatrix}.$$

Step 2:

$$\mathrm{tr}(A^{-1}) = \frac{2t}{t^2 - 1} = 1 \Rightarrow t^2 - 2t - 1 = 0.$$

Step 3:

$$t = 1 \pm \sqrt{2}.$$

Neither value is 1 or -1, so both give valid inverses.

Worked derivation:

$$(t-1)^2 = 2 \Rightarrow t = 1 \pm \sqrt{2}, \qquad t^2 - 1 = 2t \neq 0.$$

Why: A trace condition on an inverse can be used only after invertibility is checked. The quadratic produces two valid branches, and both must survive the determinant restriction.

Why not the others: (B) and (D) each make the determinant zero; (C) consists entirely of singular cases.

Key point: Keep the nonzero-determinant condition while solving an inverse constraint.

Q38 / B without-replacement conditional even product

Answer: (B) 1/7.

Given: There are three odd tickets and two even tickets.

Formula/Principle: Count pairs with even product by excluding the pairs of two odd tickets.

Step 1: Of C(5,2)=10 pairs, C(3,2)=3 have odd product; exactly C(2,2)=1 pair has both even.

Worked derivation:

$$P(\text{both even} \mid \text{product even}) = \frac{\binom{2}{2}}{\binom{5}{2} - \binom{3}{2}} = \frac{1}{10-3} = \frac{1}{7}.$$

Why: The observation rules out the three odd-odd pairs but leaves mixed-parity pairs in the conditional sample space. Counting only the even-even and odd-odd pairs would incorrectly exclude six valid even-product outcomes.

Why not the others: (A) uses the unconditional denominator; (C) counts an even first ticket but not the pair condition; (D) assumes only even-even and odd-odd outcomes.

Key point: Rebuild the conditioned sample space.

Q39 / C identical-item grouping

Answer: (C) 15.

Given: Count solutions $n_1+n_2+n_3=4$ with $n_i \geq 0$.

Formula/Principle: Stars and bars gives C(4+3−1,3−1).

Step 1: C(6,2)=15.

Worked derivation:

$$N = \binom{6}{2} = \frac{6 \cdot 5}{2} = 15.$$

Why: Identical items within each type make order irrelevant, so permutations of the same count triple are not different choices. Size four already guarantees nonempty.

Why not the others: (A) misses allocations; (B) uses fewer stars; (D) removes one valid triple.

Key point: Count type multiplicities, not sequences.

Q40 / D inverse constant-term coefficient parameter

Answer: (D) 5.

Given: A constant term choosing k factors x^{-2} has coefficient C(6,k)=15.

Formula/Principle: The coefficient condition permits k=2 or k=4; the exponent condition is p(6−k)−2k=0.

Step 1: k=2 gives p=1, while k=4 gives p=4; add the two admissible positive values.

Worked derivation:

$$\binom{6}{k} = 15 \Rightarrow k \in \{2,4\}; \quad p = \frac{2k}{6-k} \Rightarrow p \in \{1,4\}, \quad \sum p = 5.$$

Why: The same binomial coefficient occurs at two symmetric term indices, so there are two separate exponent equations to check. Selecting only one occurrence of coefficient 15 would miss one valid parameter.

Why not the others: (A) and (B) report individual parameters, while (C) adds the two term indices rather than the parameters.

Key point: Solve coefficient and exponent conditions together.

Q41 / A Positive GP terms from sum and product

Answer: (A) 121.

Given: $b^2 = ac,\ a+b+c = 13,\ abc = 27.$

Formula/Principle: The middle term of three consecutive positive GP terms is their geometric mean.

Step 1:

$$abc = b^3 = 27 \Rightarrow b = 3.$$

Step 2:

$$a + c = 10, \quad ac = 9 \Rightarrow (a, c) = (1, 9)$$

because the terms increase.

Step 3: The common ratio is $r = b/a = 3$.

Step 4:

$$S_5 = a\frac{r^5 - 1}{r - 1} = \frac{243 - 1}{2} = 121.$$

Worked derivation:

$$1 + 3 + 9 + 27 + 81 = 121.$$

Why: Positivity makes the cube-root choice unique, and strict increase determines which outer root is the first term. This is a finite geometric-sum problem and needs no test for convergence of a general infinite series.

Why not the others: (B) sums only four terms; (C) drops the initial one; (D) reports the next power rather than the sum.

Key point: Use the product to fix the middle GP term, then recover order and ratio.

Q42 / B common-root condition

Answer: (B) 9.

Given: First polynomial factors as (x−1)(x−2), with roots 1 and 2.

Formula/Principle: A common root r satisfies r²−kr+4=0, so k=r+4/r.

Step 1: r=1 gives k=5; r=2 gives k=4; sum=9.

Worked derivation:

$$k_1 = 1 + 4 = 5, \quad k_2 = 2 + 2 = 4, \quad k_1 + k_2 = 9.$$

Why: Both roots of the first polynomial can independently be common roots for different parameter values. The question asks for the sum of those possible parameters rather than either single one.

Why not the others: (A) counts only the first possibility; (C) mistakes one parameter as 2; (D) adds neither correct pair.

Key point: Check each candidate common root.

Q43 / C affine variance transform

Answer: (C) 36.

Given: Adding constant 5 does not change deviations; multiplying by 2 doubles each deviation.

Formula/Principle: Var(aX+b)=a²Var(X).

Step 1: 2²×9=36.

Worked derivation:

$$\mathrm{Var}(2X + 5) = 4\,\mathrm{Var}(X) = 4(9) = 36.$$

Why: Variance averages squared deviations from the transformed mean, so scaling appears squared and translation cancels. Confusing standard deviation with variance would give 18.

Why not the others: (A) adds scales; (B) multiplies by 2 only; (D) squares the translated value.

Key point: Scale variance quadratically.

Q44 / C formation from curve family

Answer: (C) y′=2y/x.

Given: Differentiate y=Cx² to obtain y′=2Cx.

Formula/Principle: From original equation C=y/x².

Step 1: y′=2(y/x²)x=2y/x.

Worked derivation:

$$y' = 2Cx = 2(y/x^2)x = 2y/x.$$

Why: Eliminating the arbitrary constant creates a relation common to the entire family, rather than a solution for one selected C. The x>0 domain makes division by x valid throughout.

Why not the others: (A) retains C; (B) misses factor 2; (D) inverts y.

Key point: Differentiate and substitute the parameter away.

Q45 / A exponential limit

Answer: (A) 2.

Given: e^(2x)=1+2x+(2x)²/2+O(x³).

Formula/Principle: Leading remaining numerator term is 2x².

Step 1: Divide by x² then take x→0.

Worked derivation:

$$\frac{e^{2x} - 1 - 2x}{x^2} = 2 + O(x) \longrightarrow 2.$$

Why: Subtracting both constant and linear terms isolates the quadratic contribution; direct substitution gives an indeterminate zero over zero. The limit is finite because higher powers vanish after division.

Why not the others: (B) drops factor 2; (C) forgets factorial; (D) substitutes directly.

Key point: Expand to the first surviving order.

Physics

Q46 / B vertical-motion turning time

Answer: (B) 2 s.

Given: v(t)=u−gt.

Formula/Principle: At a smooth turning point vertical velocity is zero.

Step 1: 20−10t=0, so t=2 s.

Worked derivation:

$$t_{\text{top}} = u/g = 20/10 = 2 \text{ s}.$$

Why: The balcony elevation changes the absolute height but not the time for upward velocity to reach zero. Gravity acts downward during the entire flight, including the turning instant.

Why not the others: (A) leaves positive speed; (C) and (D) occur after turning.

Key point: Solve velocity zero, not displacement zero.

Q47 / C coupled tension with support acceleration

Answer: (C) 36 N.

Given: Mass has upward acceleration 2 relative to ground.

Formula/Principle: T−mg=ma.

Step 1: T=3(10+2)=36 N.

Worked derivation:

$$T = m(g + a) = 3(12) = 36 \text{ N}.$$

Why: Stationary relative to the accelerating cabin does not mean zero acceleration in an inertial frame. The cord must both balance gravity and supply net upward force.

Why not the others: (A) uses g−a; (B) ignores acceleration; (D) overadds it.

Key point: Specify the reference frame for acceleration.

Q48 / D position-dependent work

Answer: (D) $2\sqrt{3}\ \text{m s}^{-1}$.

Given: $m = 2$ kg, $v_0 = 2\ \text{m s}^{-1}$; the graph rises as $F = 2x$ to 2 m and stays at 4 N to 3 m.

Formula/Principle:

$$K_f - K_i = W = \int F\,dx.$$

Step 1:

$$K_i = \tfrac{1}{2}(2)(2)^2 = 4 \text{ J}.$$

Step 2: The two work contributions are

$$W_1 = \int_0^2 2x\,dx = 4 \text{ J}, \quad W_2 = \int_2^3 4\,dx = 4 \text{ J}.$$

Step 3:

$$K_f = 4 + 4 + 4 = 12 \text{ J}.$$

Step 4:

$$\tfrac{1}{2}(2)v_f^2 = 12 \Rightarrow v_f = \sqrt{12} = 2\sqrt{3}\ \text{m s}^{-1}.$$

Worked derivation:

$$v_f = \sqrt{v_0^2 + 2(W_1 + W_2)/m} = 2\sqrt{3}\ \text{m s}^{-1}.$$

Why: The graph encodes force over position, so its signed area is work rather than momentum change. Initial kinetic energy must be included before converting final energy back to speed.

Why not the others: (A) ignores work; (B) treats work alone as final kinetic energy; (C) incorrectly adds an energy increment directly to speed.

Key point: Integrate force, add initial energy, then take the physical speed root.

Q49 / A rolling energy partition

Answer: (A) 1/3.

Given: I=½mR² and ω=v/R.

Formula/Principle: Krot=½Iω²=¼mv²; Ktrans=½mv².

Step 1: Krot/Ktotal=(1/4)/(3/4)=1/3.

Worked derivation:

$$\frac{K_{\text{rot}}}{K_{\text{total}}} = \frac{\frac{1}{4}mv^2}{\frac{3}{4}mv^2} = \frac{1}{3}.$$

Why: Static friction allows rolling without dissipation, but kinetic energy splits between translation and rotation. Using mgh alone without partition would miss the required fraction.

Why not the others: (B) uses I ratio directly; (C) ignores rotational energy; (D) reverses the partition.

Key point: Combine the two kinetic contributions.

Q50 / B escape speed ratio

Answer: (B) 1.

Given: vesc=√(2GM/R).

Formula/Principle: Ratio √[(2M/2R)/(M/R)].

Step 1: Ratio=1.

Worked derivation:

$$\frac{v_{e,p}}{v_{e,E}} = \sqrt{\frac{M_p/R_p}{M_E/R_E}} = \sqrt{\frac{2}{2}} = 1.$$

Why: Doubling gravitational mass strengthens attraction while doubling launch radius weakens it by the same factor in the energy ratio. Surface gravity itself need not stay equal.

Why not the others: (A) scales radius alone; (C) mass alone; (D) omits square root.

Key point: Escape speed scales with √(M/R).

Q51 / C buoyancy under partial immersion

Answer: (C) 0.6.

Given: Weight=ρblock gV; buoyancy=ρwater gVsub.

Formula/Principle: Floating balance gives Vsub/V=ρblock/ρwater.

Step 1: 600/1000=0.6; uniform cross section gives same height fraction.

Worked derivation:

$$h_{\text{sub}}/h = V_{\text{sub}}/V = 600/1000 = 0.6.$$

Why: The fluid displaces only the submerged volume, not the whole block. The density ratio determines immersion for a freely floating object before any external force is added.

Why not the others: (A) gives emergent fraction; (B) assumes half; (D) implies neutral or denser block.

Key point: Balance displaced-water weight with block weight.

Q52 / D first-law sign convention

Answer: (D) +30 J.

Worked derivation:

$$\Delta U = Q - W = 50\ \text{J} - 20\ \text{J} = +30\ \text{J}.$$

Why: Heat entering the gas increases internal energy, while work done by the gas transfers part of that input out. The explicit work sign convention prevents the opposite sign choice.

Why not the others: (A) reverses both terms; (B) reverses net sign; (C) adds rather than subtracts work.

Key point: Define work sign before applying first law.

Q53 / A field cancellation locus

Answer: (A) Midpoint.

Worked derivation:

$$E_{-a}(0) = \frac{kq}{a^2}\hat{x}, \quad E_{+a}(0) = -\frac{kq}{a^2}\hat{x} \Rightarrow E_{\text{net}}(0) = 0.$$

Why: Between like charges the fields point oppositely and equal magnitudes require equal distances, hence the midpoint. Outside the two charges both fields point the same way and cannot cancel.

Why not the others: (B) and (C) lie outside with co-directed fields; (D) overlooks symmetry.

Key point: Check direction before equating magnitudes.

Q54 / B dielectric with battery disconnected

Answer: (B) 1/4.

Given: Disconnection fixes Q; dielectric raises C to 4C.

Formula/Principle: U=Q²/(2C).

Step 1: Uf/Ui=C/Cf=1/4.

Worked derivation:

$$\frac{U_f}{U_i} = \frac{Q^2/(2 \cdot 4C)}{Q^2/(2C)} = \frac{1}{4}.$$

Why: With no battery pathway, charge cannot adjust while dielectric polarisation raises capacitance. The lost field energy corresponds to work associated with dielectric insertion, so fixed-voltage reasoning would answer another setup.

Why not the others: (A) reverses ratio; (C) squares it; (D) uses C².

Key point: Identify isolated versus battery-connected boundary.

Q55 / C force with vector field components

Answer: (C) +z, 6 N.

Given: v perpendicular to B.

Formula/Principle: F=qv×B, magnitude qvB sin90°.

Step 1: i×j=k, 1×2×3=6 N.

Worked derivation:

$$\mathbf{F} = (1)(2\mathbf{i}) \times (3\mathbf{j}) = 6\mathbf{k} \text{ N}.$$

Why: The magnetic force is perpendicular to both vectors and the positive charge follows the right-hand orientation. Reversing charge sign or vector order would reverse direction without changing magnitude.

Why not the others: (A) reverses sign; (B) misses speed; (D) chooses field direction.

Key point: Cross-product order fixes force sign.

Q56 / D inductor energy from current

Answer: (D) 9 J.

Given: L=2 H, I=3 A.

Formula/Principle: U=½LI².

Step 1: ½×2×9=9 J.

Worked derivation:

$$U = \frac{1}{2}(2)(3)^2 = 9 \text{ J}.$$

Why: Current appears squared because building the field requires work against a growing self-induced emf. Inductance alone or current alone does not determine energy.

Why not the others: (A) omits square; (B) omits half; (C) mishandles both.

Key point: Square current before multiplication.

Q57 / A real-virtual image sign

Answer: (A) Virtual upright at 10 cm left.

Given: f=+10 cm, u=−5 cm.

Formula/Principle: 1/f=1/v−1/u.

Step 1: 1/v=1/10−1/5=−1/10, so v=−10 cm; magnification v/u=+2.

Worked derivation:

$$v = -10 \text{ cm}, \quad m = v/u = (-10)/(-5) = +2.$$

Why: An object inside the focal length produces outgoing diverging rays whose backward extensions meet on the object side. A negative image distance and positive magnification together identify virtual upright imaging.

Why not the others: (B) reverses distance sign; (C) mismatches sign and orientation; (D) calls a negative-v image real.

Key point: Interpret both v sign and magnification.

Q58 / B diffraction central width

Answer: (B) Doubles.

Given: First minima approximately at sinθ=±λ/a.

Formula/Principle: Central linear width≈2Dλ/a at small angles.

Step 1: Replace a with a/2; width doubles.

Worked derivation:

$$w' = \frac{2D\lambda}{a/2} = 2w.$$

Why: Narrowing the aperture spreads the diffraction pattern because the first zero shifts farther from centre in angle. The two sides of central lobe both scale the same way.

Why not the others: (A) treats width proportional to a; (C) squares effect; (D) ignores slit.

Key point: Diffraction width is inverse to slit width.

Q59 / C line-series limit

Answer: (C) Discrete emission lines.

Worked derivation:

$$\Delta E = h\nu = hc/\lambda \quad \text{for allowed atomic transitions.}$$

Why: Isolated excited atoms emit photons at differences between their allowed energy levels, yielding characteristic lines. A dense thermal source blends many processes into a more continuous spectrum under the usual conditions.

Why not the others: (A) ignores discrete levels; (B) denies emission; (D) ignores element-specific transitions.

Key point: Relate line positions to atomic energy gaps.

Q60 / D Infer half-life then apply source-only attenuation and background

Answer: (D) 150 counts min^{-1}.

Given: The two initial observations use the same source detection efficiency; only the final screen changes detected source events.

Formula/Principle: Subtract a constant background before applying exponential decay, then apply any source-only attenuation before restoring the background.

Step 1: Source rates are $900 - 100 = 800$ initially and $500 - 100 = 400$ after 4 h, establishing a half-life of 4 h.

Step 2: At 12 h, three half-lives have elapsed:

$$R_s = 800(1/2)^3 = 100 \text{ min}^{-1}.$$

Step 3: The screen reduces that source contribution to $100/2 = 50$ min^{-1}.

Step 4:

$$R_{\text{total}} = 50 + 100 = 150 \text{ min}^{-1}.$$

Worked derivation:

$$R_{\text{total}}(12) = 100 + \tfrac{1}{2}(900 - 100)2^{-12/4} = 150 \text{ min}^{-1}.$$

Why: The independent electronic baseline neither decays with the source nor passes through the source-attenuating screen. Separating those contributions is necessary both when inferring the half-life and when predicting the final detector reading.

Why not the others: (A) loses the surviving source term; (B) ignores the screen; (C) miscalculates the number of elapsed half-lives or restores background more than once.

Key point: Infer decay from background-corrected data before changing the detection model.